

# LAKSHMI MANASA MADDI

San Francisco Bay Area, CA

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## Education

University of California, San Diego [UCSD]

Sept. 2023 – Mar. 2027

Bachelor of Science in Data Science (Concentration: AI/ML), Minor: Business Analytics

San Diego, CA

- **Awards: 1st Place at JPMorgan Chase Data for Good Hackathon** — made an educational analytics ML tool,
- **2nd Place for Data Science Student Society Annual Showcase** — built a time series + ML forecasting model,
- **Top 10 for WIC Programming Competition (100+ teams)** — solved DS/Algorithm challenges in Python + Java,
- **1st Place for SDSU Big Data Hackathon (Geocomputational Thinker Award)** — built GIS-based ML app.

## Experience

IBM (International Business Machines)

Jun. 2025 – Aug. 2025

Artificial Intelligence Architect Intern — IBM National Market

Dallas, TX

- Engineered **Python- and SQL-based data pipelines** using LangChain and Elasticsearch to unify **100K+ customer records**, achieving **95%+ accuracy** and enabling dashboards used to analyze onboarding and retention trends.
- Built a **Generative AI RAG agent** in Watsonx Orchestrate to automate insight retrieval across Salesforce and Slack, reducing **manual analysis time by 70%** and improving team efficiency by **15+ hours per week**.
- Conducted **churn and renewal analysis** using hypothesis testing and time series modeling, identifying a **12% higher churn rate** for delayed onboardings and translating findings into changes to product adoption strategy.

Halicioğlu Data Science Institute (HDSI) Tilos AI Lab

Jan. 2025 – Jun. 2025

Undergraduate Data Science Research Fellow

San Diego, CA

- Built **predictive classification models** on wearable sensor time-series data using **KNN** and **decision trees** in **scikit-learn**, achieving **91% accuracy** and identifying patterns through iterative model evaluation.
- Developed reproducible workflows with **Python** for data preprocessing and **SQL** for structured extraction, fine-tuned **transformer models** for sequence prediction, and automated **Tableau dashboards** to reduce reporting time by **38%**.

Wild Genomics

Jul. 2024 – Sep. 2024

Data Science Intern

San Diego, CA

- Optimized large-scale genomic data workflows on **AWS Spark**, processing **50K+ DNA samples** and reducing runtime by **30%**, enabling faster iteration and more timely analytical insights.
- Applied **regression, clustering, and anomaly detection** to genomic datasets to identify patterns and sequencing errors, improving model accuracy by **18%** and validating results through rigorous **SQL-based checks**.
- Built interactive **Power BI, Looker, and AWS QuickSight dashboards** to surface key metrics and trends, reducing manual reporting by **50%** and communicating insights that supported data-driven research decisions.

## Projects

Modeling Economic Trajectories Analysis | Python, XGBoost, Time Series, SQL, Tableau

Jun. 2025

- Built **hybrid ML and time series models (XGBoost, SARIMA)** on global economic data, achieving  **$R^2 = 0.96$**  and outperforming baseline forecasts by over **150%**.
- Identified key growth drivers using **lag features, rolling statistics, and SHAP analysis**; developed reproducible workflows in **Python** and **SQL**, and built interactive **Tableau dashboards** to communicate insights. [Project Link](#)

National Power Outage Duration Predictor | Python, SQL, MATLAB, Tableau

Apr. 2025

- Analyzed **1,500+ U.S. outage events** with **regression** and **anomaly detection models** in **Python** and **MATLAB**, improving prediction accuracy by **12%**.
- Streamlined data processing with **SQL** and **Python**, and deployed automated **Tableau dashboards** that reduced reporting time by **30%** and improved decision-making speed. [Project Link](#)

## Technical Skills

**Programming & Data Systems:** Python (Pandas, NumPy, Scikit-learn, LangChain), R, SQL, PostgreSQL, Java

**Machine Learning & AI:** Regression, Classification, Forecasting, Clustering, Decision Trees

**Analytics & Experimentation:** A/B Testing, Hypothesis Testing, Experiment Design, Causal Inference, Product & Service Telemetry, Data Exploration, Interpretation & Validation, Monitoring & Iteration, Econometrics

**Statistics & Operations Research:** Probability, Sampling, Inference, Optimization, Simulation, Time Series Analysis

**Visualization & Reporting:** Tableau, Power BI, Looker, Matplotlib